

Bank Account Management System

Sample Case Study on Classes & Objects – CO1

1. Aim

To design and implement a Bank Account Management System using Java classes and objects. The program demonstrates classes, objects, inheritance, constructors, method overriding, deposit, withdrawal, interest calculation, and transfer operations.

2. Problem Statement

Design a system to manage bank accounts using Account, SavingsAccount, and CurrentAccount classes. The system should store account information and perform transactions such as deposit, withdrawal, interest calculation, and transfer between accounts.

3. Classes and Objects

Class	Attributes	Methods / Purpose
Account	accnum, balance, acctype	details(), deposit(), withdraw(), transfer()
SavingsAccount	interstrate	calculateintrest(), account(); inherits Account
CurrentAccount	overdraftlimit	withdraw() overridden, account(); inherits Account
Bank	No instance attributes	main() creates objects and performs transactions

4. Algorithm

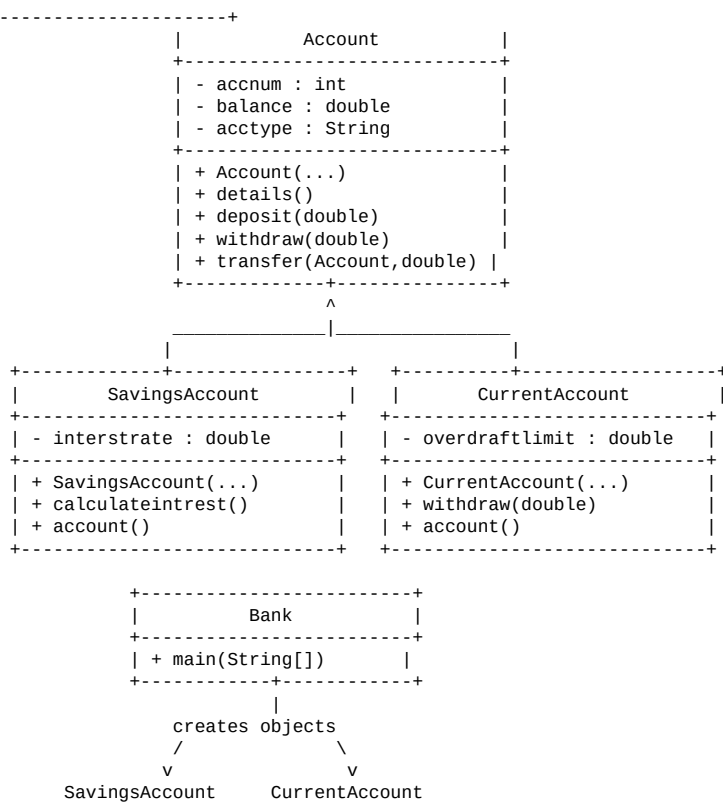
1. Start the program.
2. Create a SavingsAccount object with account number 2000, balance 2424, and interest rate 5%.
3. Create a CurrentAccount object with account number 2154, balance 5845, and overdraft limit 500.
4. Display the initial details of both accounts.
5. Deposit 1000 into the savings account.
6. Withdraw 2000 from the current account.
7. Calculate 5% interest on the savings account and add it to the balance.
8. Transfer 500 from the savings account to the current account.
9. Withdraw 5000 from the current account using its overdraft facility.
10. Display the final details of both accounts.
11. Stop the program.

5. Execution Steps

1. Open Eclipse IDE and create a Java project.
2. Create the package Excersice.
3. Create the file Bank.java.
4. Write the Account, SavingsAccount, CurrentAccount, and Bank classes.
5. Save and compile the program.

6. Run the Bank class.
7. Observe each transaction in the console.
8. Verify the final account balances.

6. UML Class Diagram



7. Dry Run / Transaction Process

Step	Operation	Savings	Current
Initial	Objects created	2424.0	5845.0
1	Deposit 1000	3424.0	5845.0
2	Withdraw 2000	3424.0	3845.0
3	Interest 5% = 171.2	3595.2	3845.0
4	Transfer 500	3095.2	4345.0
5	Withdraw 5000	3095.2	-655.0

The final CurrentAccount balance is -655 because the program allows withdrawal up to balance + overdraft limit. Before the final withdrawal, the current balance is 4345 and the overdraft limit is 500, so 5000 is permitted.

8. Complete Java Program

```
package Excercise;

class Account {
    int accnum;
    double balance;
    String acctype;

    Account(int n, double m, String l) {
        this.accnum = n;
        this.balance = m;
        this.acctype = l;
    }

    void details() {
        System.out.println("accnum :" + accnum);
        System.out.println("balance :" + balance);
        System.out.println("acctype :" + acctype);
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance = balance + amount;
            System.out.println("Deposited: " + amount);
        } else {
            System.out.println("Invalid deposit amount");
        }
    }

    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient balance");
        }
    }

    void transfer(Account receiver, double amount) {
        if (amount > 0 && amount <= balance) {
            this.balance = this.balance - amount;
            receiver.balance = receiver.balance + amount;
            System.out.println("Transferred: " + amount);
            System.out.println("To Account: " + receiver.accnum);
        } else {
            System.out.println("Transfer failed");
        }
    }
}

class SavingsAccount extends Account {
    double interstrate;

    SavingsAccount(int n, double m, double k) {
        super(n, m, "saving account");
        this.interstrate = k;
    }

    void calculateintrest() {
        double intrest = balance * (interstrate / 100);
        balance = balance + intrest;
        System.out.println("value :" + intrest);
    }
}
```

```

    }

    void account() {
        System.out.println("accnum :" + accnum);
        System.out.println("balance :" + balance);
        System.out.println("interstrate :" + interstrate);
    }
}

class CurrentAccount extends Account {
    double overdraftlimit;

    CurrentAccount(int n, double m, double o) {
        super(n, m, "current account");
        this.overdraftlimit = o;
    }

    @Override
    void withdraw(double amount) {
        if (amount > 0 && amount <= balance + overdraftlimit) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("exceed overdraft limit");
        }
    }

    void account() {
        System.out.println("accnum :" + accnum);
        System.out.println("balance :" + balance);
        System.out.println("overdraftlimit :" + overdraftlimit);
    }
}

public class Bank {
    public static void main(String[] args) {
        SavingsAccount s = new SavingsAccount(2000, 2424, 5);
        CurrentAccount c = new CurrentAccount(2154, 5845, 500);

        System.out.println("savings account");
        s.account();

        System.out.println("\ncurrent account");
        c.account();

        System.out.println("\nDeposit");
        s.deposit(1000);

        System.out.println("\nwithdraw");
        c.withdraw(2000);

        System.out.println("\ninterstrate");
        s.calculateintrest();

        System.out.println("\ntransfer");
        s.transfer(c, 500);

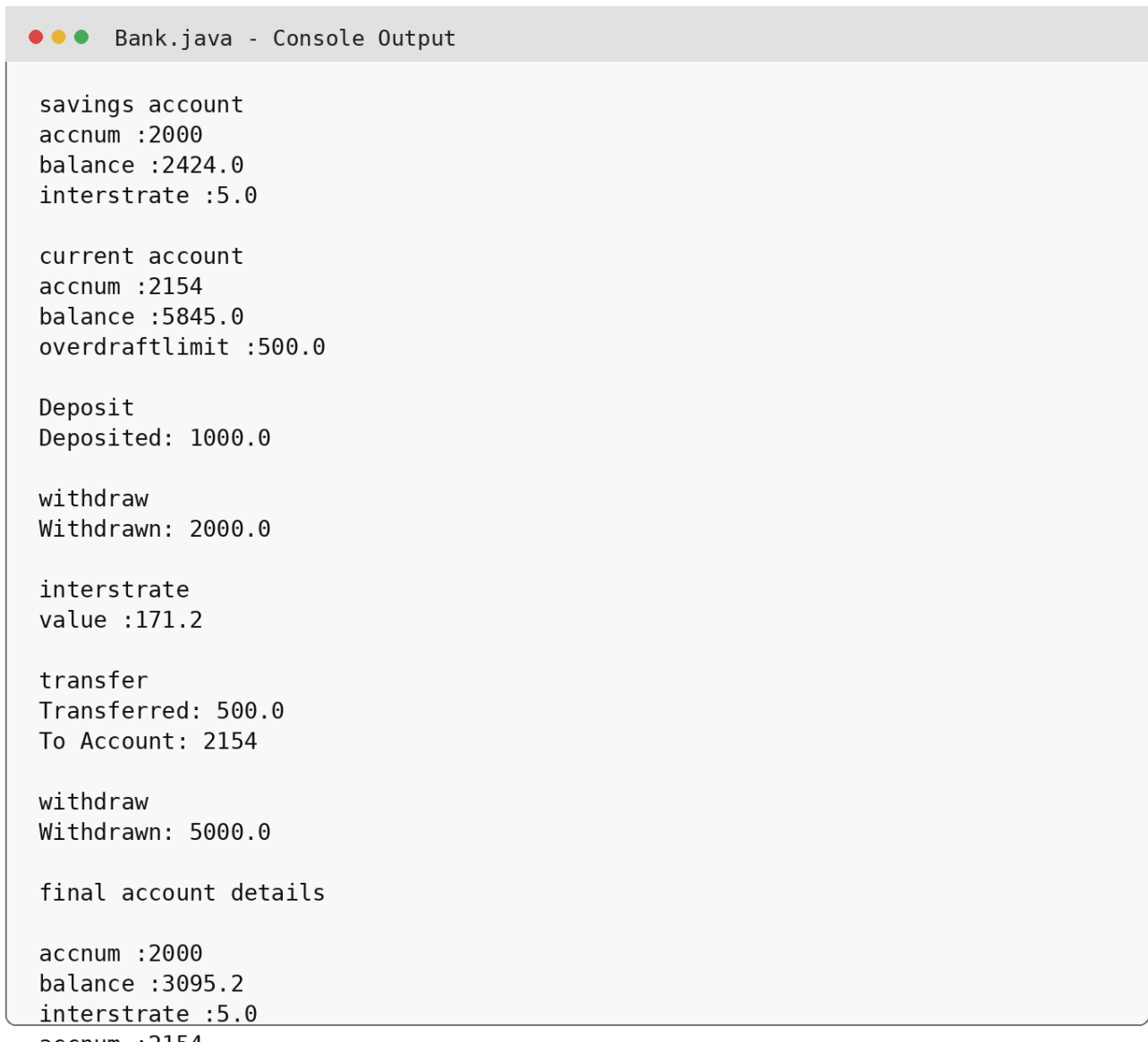
        System.out.println("\nwithdraw");
        c.withdraw(5000);

        System.out.println("\nfinal account details\n");
        s.account();
        c.account();
    }
}

```

9. Program Output – Complete Screenshot

The following screenshot represents the complete console execution of the given Java program, including account creation, deposit, withdrawal, interest calculation, transfer, final withdrawal, and final account details.



```
Bank.java - Console Output

savings account
accnum :2000
balance :2424.0
interstrate :5.0

current account
accnum :2154
balance :5845.0
overdraftlimit :500.0

Deposit
Deposited: 1000.0

withdraw
Withdrawn: 2000.0

interstrate
value :171.2

transfer
Transferred: 500.0
To Account: 2154

withdraw
Withdrawn: 5000.0

final account details

accnum :2000
balance :3095.2
interstrate :5.0
accnum :2154
```

10. Result / Conclusion

The Bank Account Management System was successfully implemented using Java classes and objects. The program demonstrates inheritance between Account and its specialized account classes, method overriding in CurrentAccount, and object interaction through banking transactions. The complete execution and final account balances are shown in the output screenshot.